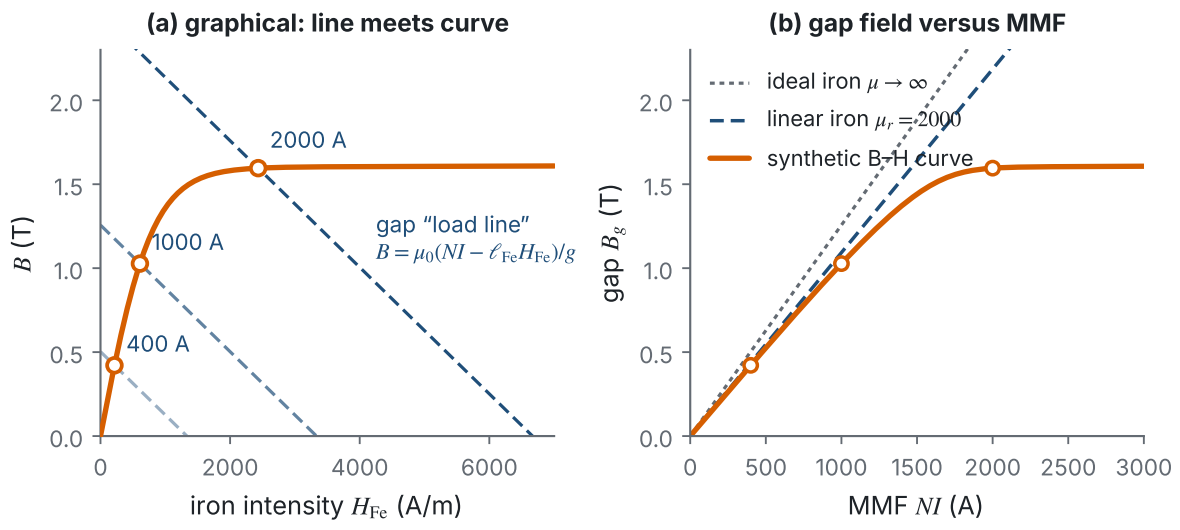
(a) graphical: line meets curve
2.0
1.5
1.0
0.5
0.0
B (T)
2000 A
1000 A
400 A
gap "load line"
$B = \mu_0(NI - \ell_{\mathrm{Fe}} H_{\mathrm{Fe}})/g$
0
2000
4000
6000
iron intensity H_{Fe} (A/m)
(b) gap field versus MMF
ideal iron $\mu \to \infty$
linear iron $\mu_r = 2000$
synthetic B–H curve
gap B_g (T)
2.0
1.5
1.0
0.5
0.0
0
500
1000
1500
2000
2500
3000
MMF NI (A)